

Abstract

Social media platforms generate rich network-structured data that, when properly analysed, can reveal sentiment dynamics, coordinated disinformation campaigns, censorship pressure, and the organic vs. artificial propagation of hashtags. This paper presents a unified methodological framework for extracting these insights from NodeXL-format social network exports. Using the Kenyan **#RejectFinanceBill2024** protest dataset (6,398 accounts, 9,806 interactions), we propose: (1) a k -means-based sentiment inference pipeline that operates even when explicit sentiment labels are absent; (2) a multi-signal disinformation scoring model combining graph-theoretic, temporal, and linguistic heuristics; (3) a censorship detection algorithm based on structural hole analysis and sudden vertex disappearance; (4) a hashtag authenticity classifier using account-level network feature vectors; and (5) a hashtag lifecycle model that tracks semantic drift and co-optation over time. All methods are designed to be computable in-browser using the SIMElab Data Explorer architecture (TypeScript/JavaScript, no backend required), making them deployable in low-resource research settings across Africa.

Keywords: social network analysis, sentiment inference, disinformation detection, censorship measurement, hashtag forensics, k -means clustering, Louvain community detection, NodeXL, Kenya elections

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1 Introduction

1.1 The Problem

The `#RejectFinanceBill2024` movement in Kenya generated thousands of Twitter/X interactions. For researchers at SIMElab Africa, the core challenge is not merely collecting this data, but extracting actionable intelligence from it. Five questions drive our inquiry:

1. **Sentiment:** What was the emotional valence of the discourse, and how did it cluster across communities?
2. **Disinformation:** Which accounts and narratives were likely inauthentic, coordinated, or deliberately misleading?
3. **Censorship:** Can we detect evidence of content suppression, account removal, or narrative manipulation?
4. **Hashtag authenticity:** Were hashtags organically propagated by real people or artificially amplified by coordinated accounts?
5. **Hashtag lifecycle:** How did a hashtag that started with legitimate grievances evolve — was it co-opted, hijacked, or diluted?

This paper proposes concrete, implementable algorithms for each question, with mathematical formulations that can be translated directly into code.

1.2 Dataset Characteristics

The `RejectFinanceBill2024` dataset is a NodeXL Pro export with the following structure:

- **Edges sheet** (9,806 rows): Vertex 1 (source handle), Vertex 2 (target handle), Relationship (Retweet/Reply/Mention/Tweet), Relationship Date (UTC), and in some exports, Tweet text.
- **Vertices sheet** (6,398 rows): Vertex (handle), Degree, In-Degree, Out-Degree, Betweenness Centrality, Closeness Centrality, Eigenvector Centrality, PageRank, Clustering Coefficient, Followers, Verified, Location, Tweets (retweet count), Favourites Count, Joined Twitter Date (UTC), and other NodeXL-computed metrics.

Critical observation: The dataset lacks explicit sentiment labels (all vertices default to “Neutral”) and tweet text is not always available in the Vertices sheet. This means we cannot rely on off-the-shelf NLP sentiment tools. We must infer sentiment from network structure and, where text is available, from lightweight lexical analysis.

1.3 Theoretical Framework

We ground our analysis in three theoretical pillars:

- **Granovetter’s Strength of Weak Ties** [1] (1973): The structure of information flow in networks reveals community boundaries. Weak ties bridge clusters; strong ties reinforce echo chambers.

- **Barabási–Albert Preferential Attachment** [2] (1999): In scale-free networks like Twitter, a few nodes accumulate disproportionate influence. These hubs are critical for both organic and artificial information spread.
- **Goffman’s Frame Analysis** [3] (1974): Hashtags function as frames — interpretive schemata that shape how events are understood. Frame contestation (competing narratives around the same hashtag) is a key mechanism for co-optation.

2 Sentiment Analysis via K -Means Clustering on Network Feature Vectors

2.1 Why Network-Based Sentiment?

When tweet text is absent or in languages poorly served by NLP models (Swahili, Sheng, vernacular Kenyan English), we cannot rely on lexical sentiment. However, **sentiment leaves structural fingerprints** in social networks:

- Accounts expressing **negative** sentiment (anger, protest) tend to have high out-degree (they broadcast), low reciprocity (they don’t engage in dialogue), and cluster in dense, like-minded communities.
- Accounts expressing **positive** sentiment (support, celebration) tend to have reciprocal relationships, higher clustering coefficients, and bridge across communities.
- **Neutral** accounts (observers, journalists) have high betweenness centrality (they connect clusters) but moderate degree.

2.2 Feature Vector Construction

For each vertex v_i , we construct a 9-dimensional feature vector $\mathbf{x}_i \in \mathbb{R}^9$:

$$\mathbf{x}_i = \begin{bmatrix} \deg_{in}(v_i) \\ \deg_{out}(v_i) \\ C_B(v_i) \\ C_C(v_i) \\ C_E(v_i) \\ PR(v_i) \\ CC(v_i) \\ r_i \\ f_i \end{bmatrix} \quad (1)$$

where:

- $C_B(v_i)$ = Betweenness Centrality (Brandes algorithm [4])
- $C_C(v_i)$ = Closeness Centrality (BFS-based)
- $C_E(v_i)$ = Eigenvector Centrality (power iteration)
- $PR(v_i)$ = PageRank (iterative, $\alpha = 0.85$)

- $CC(v_i)$ = Clustering Coefficient (Watts–Strogatz local [6])
- r_i = Reciprocity ratio = $\frac{\text{mutual edges}}{\text{total edges for } v_i}$
- f_i = Follower-to-following ratio (normalized)

All features are normalized to $[0, 1]$ via min-max scaling:

$$\hat{x}_j = \frac{x_j - \min(X_j)}{\max(X_j) - \min(X_j)} \quad (2)$$

2.3 K -Means Clustering Algorithm

Algorithm 1: Network-Feature Sentiment Clustering

Input : Set of vertices V , feature vectors $X = \{x_1, \dots, x_n\}$, $k = 3$

Output : Cluster assignments $C = \{c_1, \dots, c_n\}$ where $c_i \in \{\text{Pos}, \text{Neu}, \text{Neg}\}$

Step 1. Initialize k centroids μ_1, μ_2, μ_3 using k -means++ [7]:

- Choose μ_1 uniformly at random from X
- For $j = 2$ to k : compute $D(x_i) = \min_{m < j} \|x_i - \mu_m\|^2$; choose $\mu_j \propto D(x_i)$

Step 2. Repeat until convergence ($\Delta < \varepsilon = 0.001$):

- Assignment*: $c_i \leftarrow \arg \min_{j \in \{1,2,3\}} \|x_i - \mu_j\|^2$
- Update*: $\mu_j \leftarrow \frac{1}{|S_j|} \sum_{x_i \in S_j} x_i$

Step 3. Post-clustering label assignment:

Compute mean out-degree $\bar{o}_j$, mean reciprocity $\bar{r}_j$, mean betweenness $\bar{b}_j$ per cluster

- $\max \bar{o}_j$ and $\min \bar{r}_j \rightarrow \mathbf{Neg}$ (angry broadcasters)
 $\max \bar{r}_j$ and $\max \bar{b}_j \rightarrow \mathbf{Pos}$ (reciprocal connectors)
 Remaining cluster $\rightarrow \mathbf{Neu}$ (neutral observers)
-

2.4 Determining Optimal K

Use the **elbow method** with the within-cluster sum of squares (WCSS):

$$WCSS(k) = \sum_{j=1}^k \sum_{x_i \in S_j} \|x_i - \mu_j\|^2 \quad (3)$$

Plot $WCSS(k)$ for $k = 1, 2, \dots, 10$. The “elbow” where the marginal reduction drops significantly indicates the natural number of clusters.

2.5 Validation Against Ground Truth

When manually labelled data exists, compute the Adjusted Rand Index:

$$\text{ARI} = \frac{RI - \mathbb{E}[RI]}{\max(RI) - \mathbb{E}[RI]} \quad (4)$$

For our dataset (no ground truth), use the **silhouette score** [8] as internal validation:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (5)$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance for point i . A mean score > 0.5 indicates well-separated clusters.

2.6 Sentiment Volatility Over Time

Once vertices are labelled, compute the sentiment time series:

$$S_{\text{sent}}(t) = \frac{|V_{\text{sent}}(t)|}{|V(t)|} \quad (6)$$

where $V_{\text{sent}}(t)$ is the set of vertices of class *sent* active at time bin t (daily bins). Sentiment shift detection uses a CUSUM chart [9]:

$$G_t = \max(0, G_{t-1} + (S_{\text{Neg}}(t) - \mu_{\text{Neg}} - \delta)) \quad (7)$$

When G_t exceeds threshold $h = 3\sigma$, a sentiment regime change is flagged.

3 Disinformation Detection

3.1 The Multi-Signal Approach

Disinformation on Twitter manifests through multiple detectable signals. No single signal is conclusive, but their combination produces a robust detection score.

3.2 Signal Definitions

3.2.1 Signal 1: Retweet Amplification Anomaly (A_1)

Coordinated disinformation relies on mass retweeting:

$$A_1(v_i) = \frac{\text{retweets}(v_i) + 1}{\text{followers}(v_i) + 1}, \quad \hat{A}_1(v_i) = \min \left(1, \frac{A_1(v_i)}{P_{95}(A_1)} \right) \quad (8)$$

3.2.2 Signal 2: Temporal Regularity (A_2)

Humans tweet irregularly; bots tweet regularly:

$$A_2(v_i) = 1 - \frac{\sigma(\Delta t_i)}{\mu(\Delta t_i)} \quad (9)$$

where Δt_i is the set of inter-tweet intervals for account v_i . Values near 1 suggest automation.

3.2.3 Signal 3: Network Position Anomaly (A_3)

Disinformation accounts broadcast but do not bridge communities:

$$A_3(v_i) = \frac{\deg_{out}(v_i) + 1}{\deg_{out}(v_i) + C_B(v_i) + 1} \quad (10)$$

Values near 1 indicate high broadcasting with no bridging.

3.2.4 Signal 4: Echo Chamber Index (A_4)

$$A_4(v_i) = \frac{|\{e_{ij} : \text{cluster}(v_j) = \text{cluster}(v_i)\}|}{\deg(v_i)} \quad (11)$$

High values indicate the account only interacts with its in-group.

3.2.5 Signal 5: Follower Graph Sparse Connectivity (A_5)

$$A_5(v_i) = 1 - CC_{\text{ego}}(v_i) \quad (12)$$

where $CC_{\text{ego}}(v_i)$ is the clustering coefficient of the ego-network. Low clustering is suspicious.

3.3 Composite Disinformation Score

$$D(v_i) = \sum_{m=1}^5 w_m \cdot \hat{A}_m(v_i), \quad \mathbf{w} = [0.25, 0.15, 0.30, 0.15, 0.15] \quad (13)$$

Classification thresholds:

$$v_i \text{ is classified as } \begin{cases} \text{Clean} & D(v_i) < 0.35 \\ \text{Suspicious} & 0.35 \leq D(v_i) < 0.60 \\ \text{Likely Disinfo} & D(v_i) \geq 0.60 \end{cases} \quad (14)$$

3.4 Coordinated Behaviour Detection

Algorithm 2: Coordination Ring Detection

Input : Edges E , time window τ (default: 300 seconds)
Output : Sets of coordinated account groups $\mathcal{G} = \{g_1, \dots, g_k\}$
foreach edge $e_{ij} = (v_i, v_j, t, \text{Retweet})$ **do**
 | Group edges by target v_j
end
foreach target v_j with retweeters $R_j = \{(v_i, t_i)\}$ **do**
 | Sort R_j by t_i ; scan with sliding window of size τ ;
 | **foreach** window W where $|W| \geq 3$ **do**
 | Compute Jaccard similarity $J = |S_W \cap S_{W'_k}| / |S_W \cup S_{W'_k}|$;
 | **if** $J > 0.5$ **then** add $S_W \cup S_{W'_k}$ as candidate coordination ring;
 | **end**
end
Merge overlapping rings (union-find);
foreach merged ring g_m **do**
 | $\text{coord}(g_m) \leftarrow |g_m| \cdot \bar{J}_m / |\text{targets}(g_m)|$;
 | **if** $\text{coord}(g_m) > 3.0$ **then** flag g_m as a coordination ring;
end

3.5 Narrative Consistency Score

$$\text{NCS}(v_i) = 1 - \frac{|\text{unique bigrams}(v_i)|}{|\text{total bigrams}(v_i)|} \quad (15)$$

When $\text{NCS}(v_i) > 0.7$, the account repeats the same phrases. Combined with high $D(v_i)$, this strongly indicates a disinformation account.

4 Censorship Detection

4.1 The Challenge

Censorship is the removal of content — by definition, it is the observation of *absence*. Detecting it requires inferring what *should* be present but is not.

4.2 Structural Hole Analysis

Algorithm 3: Structural Hole Censorship Detection

Input : Two network snapshots G_{t_1}, G_{t_2} (e.g. consecutive days)
Output : Candidate censored vertices V_{censored}
 Compute $C_B^{t_1}(v)$ for all $v \in V_{t_1}$;
 Identify missing vertices $V_m = V_{t_1} \setminus V_{t_2}$;
foreach $v_m \in V_m$ **do**
 $SI(v_m) \leftarrow C_B^{t_1}(v_m) \cdot \log(\deg_{t_1}(v_m) + 1)$
end
 Rank V_m by SI descending; take top- K ($K = \min(20, |V_m|)$);
foreach top- K vertex v_m **do**
 Test connectivity: if # components in $G_{t_1} \setminus \{v_m\}$ increases $> 10\%$:
 flag v_m as “censored — structural bridge removed”;
 Cross-reference Twitter API status (suspended / deleted / shadow-banned)
end

4.3 Sudden Drop Detection in Network Metrics

$$\Delta_{\text{density}}(t) = \frac{\text{density}(G_t) - \text{density}(G_{t-1})}{\text{density}(G_{t-1})} \quad (16)$$

Flag days where $\Delta_{\text{density}}(t) < -0.15$ as potential censorship events.

4.4 Hashtag Shadow-Ban Detection

$$\mathbb{E}[\text{volume}(h, t)] = \alpha \cdot \text{volume}(h, t-1) + \beta \cdot \text{mentions}(h, t) + \gamma \quad (17)$$

When the observed volume is significantly below expectation:

$$z = \frac{\text{observed} - \mathbb{E}[\text{volume}]}{\sigma_{\text{residual}}} < -2.0 \quad (18)$$

flag the hashtag as a potential shadow-ban.

5 Hashtag Authenticity Analysis

5.1 Legitimacy Feature Vector

For each account v_i using hashtag h , construct:

$$\mathbf{L}_i = \begin{bmatrix} \log(\text{followers}(v_i) + 1) \\ \text{Verified} \in \{0, 1\} \\ \text{Account age (days)} \\ \text{Tweet diversity} = |\text{unique hashtags}| / \text{total tweets} \\ \text{Engagement ratio} = \text{favorites} / (\text{followers} + 1) \\ \text{Community integration} = |\text{distinct clusters connected}| / (\text{deg}(v_i) + 1) \\ \text{Original content ratio} = \text{original tweets} / (\text{total tweets} + 1) \end{bmatrix} \quad (19)$$

5.2 Legitimacy Classifier (GMM)

Use a 2-component Gaussian Mixture Model fitted via Expectation-Maximization:

$$p(\mathbf{L}_i) = \pi_1 \cdot \mathcal{N}(\mathbf{L}_i \mid \mu_1, \Sigma_1) + \pi_2 \cdot \mathcal{N}(\mathbf{L}_i \mid \mu_2, \Sigma_2) \quad (20)$$

Algorithm 4: Hashtag Authenticity Decomposition

Input : Hashtag h , vertices V_h (accounts using h)
Output : Legitimate ratio R_h , artificial amplification score A_h
 Extract $\mathbf{L}_i$ for all $v_i \in V_h$;
 Fit 2-component GMM on $\{\mathbf{L}_i\}$;
 Label component with higher mean account age $\rightarrow$ **Legit**;
 $R_h \leftarrow |V_{\text{legit}}| / |V_h|$; $A_h \leftarrow 1 - R_h$;
if $A_h > 0.40$ **then** flag hashtag as “significantly artificially amplified”;
foreach *artificial-classified* v_i **do**
 | $\text{amp}(v_i) \leftarrow \text{retweets}(v_i) \cdot \mathbf{1}[v_i \in V_{\text{artificial}}]$
end
 Cross-reference with $D(v_i)$:
 $D(v_i) > 0.5$ and $v_i \in V_{\text{artificial}} \rightarrow$ *coordinated disinfo amplifier*
 $D(v_i) < 0.35$ and $v_i \in V_{\text{artificial}} \rightarrow$ *bot-like but not disinfo*

5.3 Hashtag Propagation Graph & Cascade Depth

$$\text{depth}(h) = \max_{v \in V_h} \text{shortest_path}(v_{\text{origin}}, v) \quad (21)$$

Organic hashtags have depth ≥ 7 ; artificial hashtags typically have depth ≤ 2 .

6 Hashtag Lifecycle Analysis

6.1 The Lifecycle State Machine

$$\text{Phase} \in \{\text{Birth}, \text{Growth}, \text{Peak}, \text{Contestation}, \text{Co-optation}, \text{Decay}, \text{Resurrection}\} \quad (22)$$

6.2 Phase Detection Algorithm

Algorithm 5: Hashtag Lifecycle Phase Detection

Input : Time series: $\text{volume}(t)$, $\text{unique_users}(t)$, $\text{sentiment}(t)$, $\text{novelty}(t)$
Output : Phase transitions $\{(t, \text{phase}_{\text{old}} \rightarrow \text{phase}_{\text{new}})\}$
BIRTH $\rightarrow$ **GROWTH**: $\text{volume}(t)$ crosses > 10 tweets/day for 2 consecutive bins;
GROWTH $\rightarrow$ **PEAK**: growth rate < 0.05 for 2 bins while volume remains high;
PEAK $\rightarrow$ **CONTESTATION**: sentiment entropy increases $> 50\%$ in one bin
AND novelty drops sharply;
CONTESTATION $\rightarrow$ **CO-OPTATION**: $\text{amplifier_ratio}(t) > 0.30$;
CONTESTATION $\rightarrow$ **DECAY**: volume $< 50\%$ of peak for 3 bins **AND**
amplifier ratio < 0.30 ;
CO-OPTATION $\rightarrow$ **DECAY**: volume $< 25\%$ of peak for 3 bins;
DECAY $\rightarrow$ **RESURRECTION**: volume crosses $> 30\%$ of peak for 2 bins;

6.3 Semantic Drift Detection

For each time window t , cluster tweet text into k topics via k -means on TF-IDF space. Track topic drift:

$$\text{drift}(t) = 1 - \max_{j,j'} \cos(\tau_j^{(t-1)}, \tau_{j'}^{(t)}) \quad (23)$$

When $\text{drift}(t) > 0.4$, the hashtag's meaning has shifted (the *semantic hijack moment*).

Compare account composition before and after t_{drift} :

$$H = \text{accounts joining after } t_{\text{drift}} \cap \text{accounts classified as artificial} \quad (24)$$

If $|H|/|\text{new accounts}| > 0.50$, the hijack was artificial.

6.4 Origin Tracing

1. v_{origin} = account with earliest timestamp for h .
2. Classify origin type:
 - **Grassroots**: degree < 100 , no verified badge, few followers.
 - **Influencer-launched**: verified, high degree, many followers.
 - **Astroturfed**: high disinformation score $D(v_{\text{origin}})$.
3. Compute adoption network density for first 50 adopters:

$$\text{organic_launch} = \frac{|\text{edges in adoption network}|}{50 \times 49} \quad (25)$$

If > 0.1 : organic launch. If < 0.02 : coordinated seeding.

7 Bad Actor & “Thief” Detection

7.1 Defining “Bad Actors”

A “bad actor” is operationally defined as an account that: exploits community trust, diverts attention to a different agenda, profiteers from crisis discourse, or infiltrates communities to sow division.

7.2 The MICE Framework (Adapted)

- **Monetization:** promotes external links, payment links in bio.
- **Infiltration:** suddenly shifts topic cluster.
- **Co-optation:** content shifts from original hashtag meaning to a competing narrative.
- **Exploitation:** amplifies crisis for follower growth, then pivots.

7.3 Infiltration Detection via Community Shift

Algorithm 6: Infiltration Detection

Input : Snapshots G_{t_1}, G_{t_2} with community assignments

Output : Infiltration candidates

foreach v_i *present in both snapshots* **do**

if $cluster_{t_2}(v_i) \neq cluster_{t_1}(v_i)$ **then**

$SA(v_i) \leftarrow |N_{t_1}(v_i) \cap N_{t_2}(v_i)| / |N_{t_1}(v_i)|$;

if $SA(v_i) < 0.2$ **then** flag v_i as “migrator”;

end

end

foreach *migrator* v_i **do**

 Test fragmentation of C_{old} with and without v_i ;

if *fragmentation increases* $> 1.5\times$ **then** flag as “community exploiter”;

if v_i *achieves top-10% centrality in C_{new} rapidly* **then** flag as “infiltrator”;

end

7.4 Engagement Farming Detection

$$\text{FarmScore}(v_i) = \frac{\text{retweets during crisis}}{\text{retweets during non-crisis}} \cdot \frac{\text{new followers during crisis}}{\text{new followers during non-crisis}} \quad (26)$$

$\text{FarmScore}(v_i) > 10$ indicates crisis farming behaviour.

8 Implementation Architecture for SIMElab Data Explorer

8.1 Current State (v5.0)

The Explorer computes: Degree, Betweenness, Closeness, Eigenvector, PageRank, Clustering Coefficient, Connected Components, Reciprocity, Density, Diameter, Louvain Commu-

nities, basic bot scoring, TF-IDF narrative detection, polarity index, and timeline event detection.

8.2 Proposed v6.0 Module Structure

```

1 src/engine/
2   sentimentClustering.ts    -- k-means on network feature
   vectors (Sec. 2)
3   disinformationScoring.ts  -- 5-signal composite score (Sec. 3)
4   coordinationDetection.ts  -- coordination ring detection (Sec.
   3.4)
5   censorshipDetection.ts    -- structural hole analysis (Sec. 4)
6   hashtagAuthenticity.ts    -- GMM legitimacy classifier (Sec. 5)
7   hashtagLifecycle.ts       -- phase detection + semantic drift
   (Sec. 6)
8   badActorDetection.ts     -- MICE framework + infiltration
   (Sec. 7)
9   types.ts                 -- Phase, Signal, LegitimacyScore,
   etc.

```

Listing 1: Proposed engine module structure

8.3 Computational Feasibility

All proposed algorithms are $O(|V|^2)$ or better.

Table 1: Per-algorithm complexity and estimated runtime at 6,398 vertices

Algorithm	Complexity	Feasible at 6,398 vertices?
<i>K</i> -means sentiment	$O(V \cdot k \cdot d \cdot \text{iter})$	Yes (~ 30 ms)
Disinfo signals 1–5	$O(V)$ each	Yes (~ 50 ms total)
Coordination rings	$O(E \log E)$	Yes (~ 100 ms)
Censorship holes	$O(V ^2)$ (Brandes BC)	Yes (~ 500 ms)
Hashtag GMM	$O(V_h \cdot \text{iter})$	Yes (~ 20 ms)
Lifecycle phases	$O(T \cdot V)$	Yes (~ 10 ms)
Infiltration	$O(V)$	Yes (~ 5 ms)
Total (in-browser, graphology.js)		< 2 seconds

9 Case Study: #RejectFinanceBill2024

9.1 Expected Network Structure

With 6,398 vertices, 9,806 edges, average degree ≈ 1.53 , and density ≈ 0.00024 , the network is **sparse and star-like** — consistent with a protest movement where a few central accounts (organizers, journalists, politicians) broadcast to many followers who retweet but rarely interact with each other.

9.2 Predicted Sentiment Clusters

- **Cluster “Neg”** (angry broadcasters): high out-degree, low reciprocity. Expected $\sim 40\%$ of vertices.
- **Cluster “Neu”** (observers/journalists): high betweenness, moderate degree, high reciprocity. Expected $\sim 50\%$.
- **Cluster “Pos”** (supporters/defenders): high reciprocity, high clustering, low out-degree. Expected $\sim 10\%$.

9.3 Predicted Disinformation Landscape

Kenyan political Twitter has documented patterns including hashtag hijacking, bot retweet rings, and engagement farming. The coordination ring detection (Algorithm 2) is expected to identify 3–8 distinct coordination rings in this dataset.

9.4 Censorship Indicators

Key indicators to watch in daily snapshots:

- Sudden disappearance of high-betweenness vertices.
- Density drops $> 15\%$ between consecutive days.
- Hashtag volume drops not explained by natural decay.

10 Limitations and Future Work

10.1 Current Limitations

1. **No tweet text in Vertices sheet:** Text-based features are unavailable and must rely on proxy signals.
2. **Cross-platform blindness:** Facebook, TikTok, and WhatsApp are not represented.
3. **No ground truth:** Human annotation of ≥ 200 accounts would significantly improve calibration.
4. **Temporal granularity:** Dates may be at day level, limiting coordination detection precision.

10.2 Future Work

1. Active data collection via Twitter/X API v2 (tweet text, precise timestamps, bio information).
2. Cross-platform integration (Facebook public pages, YouTube comments).
3. Longitudinal analysis across subsequent Kenyan hashtag movements.
4. Swahili/Sheng NLP lexicon for Kenyan political discourse.
5. Real-time monitoring pipeline with automated daily briefings.

11 Conclusion

This paper has presented a unified framework for extracting sentiment, disinformation, censorship, hashtag authenticity, and lifecycle intelligence from NodeXL-format social network data. The methods are designed to work with the data that SIMElab actually has — network structure and account metadata, even when text is sparse. They are computationally feasible in-browser and can be integrated into the SIMElab Data Explorer v6.0.

The key insight is that **network structure is a first-class signal** for answering questions traditionally approached through NLP alone. Who you retweet, how often, whether people retweet you back, what communities you bridge — these structural footprints reveal sentiment, authenticity, and intent in ways that text analysis alone cannot, especially in multilingual, code-switched environments like Kenyan Twitter.

12 Deep Mathematical Interpretation: What the Numbers Actually Mean

12.1 Interpreting Vector Angles

When accounts are represented as feature vectors in $\mathbb{R}^9$, cosine similarity reveals behavioural relationships:

- $\cos(\theta_{ij}) > 0.95$: Accounts have near-identical network behaviour. If they belong to *different* Louvain communities, this is the strongest signal of coordination — likely sock puppet accounts or coordinated bot farm members.
- **Bimodal cosine similarity distribution** (peaks near 0.3 and 1.0): Fit a 2-component GMM to the pairwise similarity histogram; the high-similarity component identifies coordination rings.
- **Hashtag self-angle** $\cos(\mathbf{h}(t_0), \mathbf{h}(t_n))$: < 0.3 means the hashtag has been hijacked; < 0 (obtuse angle) means *antithetical co-optation* — the hashtag now means the opposite of what it started as.

12.2 Spectral Analysis (Eigenvalues)

- λ_1 (**spectral radius**): Large $\Rightarrow$ influencer-dominated discourse. Moderate $\Rightarrow$ grassroots. If $\lambda_1 > 0.7 \cdot \max(\deg)$, the hashtag is celebrity/politician-driven.
- $\lambda_2(L)$ (**Fiedler value / algebraic connectivity**): Near 0 $\Rightarrow$ network is one edge away from fragmentation. The **Censorship Vulnerability Index** $= 1/\lambda_2(L) \cdot \max C_B(v)$.
- **Eigenvalue spacing distribution**: Artificially constructed bot networks may show spacing that is *too uniform* — closer to Wigner–Dyson than real human community structure.

12.3 Information-Theoretic Measures

- **Behavioural entropy** $H(v_i)$: Near 0 $\Rightarrow$ bot-like (pure retweeter). Near $\log_2 4 = 2.0 \Rightarrow$ human-like.
- **Transfer entropy** $T_{A \rightarrow B}$ [13]: Asymmetric high values indicate Community A is *driving* Community B's narrative.
- **Graph smoothness** $S(D)$: $S(D) < 0.1$ means disinformation spreads through network ties — itself evidence of coordination.

12.4 Topological Data Analysis (Persistent Homology)

- β_0 **persistence curve** [14]: Organic hashtags show slow, gradual merging of components. Artificial campaigns show sudden collapse.
- **Birth-death diagrams**: Few points far from the diagonal $\Rightarrow$ lack of genuine community structure $\Rightarrow$ artificial campaign.

12.5 K-Means Centroid Dynamics

- $\|\mu_{\text{Neg}} - \mu_{\text{Pos}}\|_2 > 3.0$: Highly polarized discourse. < 1.0 : Sentiment clustering from network structure is unreliable.
- μ_{Neu} moving toward μ_{Neg} : Discourse radicalizing (journalists taking sides).

12.6 Distance-From-Origin Test

Plot $d_{\text{cos}}(v_i, v_0)$ for each adopter sorted by join time:

- **Gradual increase**: Organic viral spread.
- **Flat curve**: Astroturfed (all accounts similarly configured).
- **Jump discontinuity**: Hashtag hijacked at that timestamp.

Practical Summary

Table 2: Interpreting key mathematical outputs

Mathematical Output	Value	Real-World Conclusion
$\cos(\theta_{ij})$ between two accounts	> 0.95	Same person or coordinated bots
$\cos(\mathbf{h}(t_0), \mathbf{h}(t_n))$	< 0.3	Hashtag meaning completely hijacked
$\lambda_2(L)$ (algebraic connectivity)	Near 0	Network fragile — one removal fragments discourse
$H(v_i)$ (behavioural entropy)	< 0.3	Bot-like behaviour
$T_{A \rightarrow B}$ (transfer entropy)	High, asymmetric	Community A manipulating Community B
β_0 persistence	Drops suddenly	Artificial/uniform network
$\ \mu_{\text{Neg}} - \mu_{\text{Pos}}\ _2$	> 3.0	Highly polarized discourse
$d_{\cos}$ curve shape	Jump discontinuity	Hashtag hijacked at that timestamp
Graph smoothness $S(D)$	< 0.1	Coordinated disinformation via network ties
Eigenvalue spacing distribution	Too uniform	Artificially generated network

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